

Direct Access and Why You Should Avoid It

As mentioned, Python does allow direct access to instance variables. [Listing 8-2](#) instantiates the same two objects from the `Person` class in [Listing 8-1](#) as in the previous section, but then accesses their `self.salary` instance variables directly.

File:

PersonGettersSettersAndDirectAccess/Main_PersonDirectAccess.py

```
# Person example main program using direct access

from Person import *

oPerson1 = Person('Joe Schmoe', 90000)
oPerson2 = Person('Jane Smith', 99000)

# Get the values of the salary variable directly
❶ print(oPerson1.salary)
   print(oPerson2.salary)

# Change the salary variable directly
❷ oPerson1.salary = 100000
   oPerson2.salary = 111111

# Get the updated salaries and print again
print(oPerson1.salary)
print(oPerson2.salary)
```

***Listing 8-2:** Example main code using direct access to an instance variable*

Python allows you to write code like this that reaches into an object to directly get ❶ and set ❷ any instance variable using standard dot syntax. Most Python programmers feel that this technique is perfectly acceptable. In fact, Guido van Rossum (the creator of Python) famously said in reference to this issue, “We are all adults here,” meaning that programmers should know what they are doing, and the risks involved, when they attempt to access instance variables directly.